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Database Systems and Data Preparation
December 10th, 2025

Final Project Proposal – Fantasy Football Database

This project will construct a database for my fantasy football league. The league is always made up of 10 managers who own teams of 14 active players. Each week, each picks their best line up (1 QB, 2 RB, 3 WRs, 1 TE, 1RB/WR/TE, 1 K and 1 DEF) and competes head-to-head. After the 14th week of the season, the postseason begins and the team with the most wins duke it out to be named champion of the league in week 17.

The league is hosted on yahoo fantasy, which has an API that tracks everything that happens in the league. The yahoo API will be the main data source since it is the only place where this information lives. This link is the documentation on the API <https://developer.yahoo.com/fantasysports/guide/>. In order to access it, I had to register as a developer on their website and link my yahoo fantasy account. Once that was set up, I was able to access all the data in every league I have ever been in. Yahoo stores the data in large JSON strings, which I will parse and transform into tabular data for SQL.

Furthermore, I will also use the library nflreadpy (docs here <https://nflreadpy.nflverse.com/>) to get some reference data. Yahoo's fantasy football API is great for tracking the activity within our league but does not do a good job of tracking NFL information like what player is on which team in a given season and what opponents they played. Nflreadpy has reference data like this which will be vital for strong modelling. What team a player is on and what team they are playing against can make the difference between starting and sitting a player in any given week.

The database will be split into two schema, manager_info, player_info, and transactions. The manager_info schema will include tables for storing the managers (the people in the league), seasons, and weekly matchups. From this we can find out who has the most wins and losses and individual records against people. The player_info schema will include tables for storing weekly manager rosters as well as a master list of players, the teams they belong to, and scheduled opponents. The transactions schema will hold tables related to trades and waiver pickups managers make within the fantasy league.

The main goal of this database is simply to record the happenings of the league with my friends, but the neural network that is trained on my specific league should be able to offer insights into which the draft and to play on a week-to-week basis. This is a very interesting project to me since it is about an activity I do with my friends and I will continue to use it for as long as the league continues to play. Thanks for the opportunity to pursue this interest in project form!

